MACHINE LEARNING BASED SUPER-RESOLUTION ALGORITHM ROBUST TO REGISTRATION ERRORS

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ABSTRACT

In this work, a novel two phase approach is proposed for robust super-resolution in the presence of registration errors and outliers. In the first phase machine learning method is used to create a weight matrix for every LR image indicating the presence of registration errors. In the second phase, super-resolution is performed using all of the LR images and the associated weight matrices, creating an image which is free of error artifacts.

Index Terms— super-resolution, image enhancement, registration error, L_2 norm minimization, logistic regression.

1. INTRODUCTION

Super-resolution algorithms are known to be very sensitive to errors in registration of the low resolution images and presence of outliers. The errors in the registration or the presence of outliers manifests as unpleasing and very annoying artifacts in the super-resolved image, causing SR procedure to degrade the image instead of enhancing it.

In presence of outliers or large errors in the alignment of the LR images the L_1 norm, proposed in [1], has proven as valuable tool, outperforming the L_2 norm. However, without the presence of outliers, especially in cases where the noise can be modeled as Gaussian, the performance of SR schemes based on L_2-norm minimization is better, giving sharper images [2]. In the work of Farsiu *et al*., [3], L_1 norm is used with the combination of Bilateral-TV regularization. This regularization is used in order to remove the noise, while keeping the sharp edges.

In the work of El-Yamany *et al*., [4], robust error function (*The Lorentzian*) is used without any regularization. Authors claim that in the presence of outliers their method is superior to the methods based on L_1 and L_2 norms. The Lorentzian influence function, which is proportional to the derivative of the Lorentzian function, assigns different weights to the error depending on the size of the error.

In the work of Takeda *et al*., [5], every pixel is approximated using 3D Taylor series. The coefficients of this series are estimated by solving a local weighted least-squares problem, where the weights are a function of the 3D space-time orientation in the neighborhood. Authors claim that with this method there is no need of explicit subpixel motion estimation.

Belekos *et al*., in [6], use multichannel image prior, instead of the usual Tikhonov regularization. A hierarchical Gaussian non-stationary prior model is created using the neighboring LR frames. The residuals of first order differences within the frame in both directions, and through time are modeled as Gaussian random variables with zero mean and spatially varying variance.

In [2] a different approach to the problem of motion estimation errors was proposed. Instead of suppressing the errors influence, selectivity was introduced, using a measure of usability of every pixel in each low-resolution image for SR enhancement. Simple techniques were proposed for fast and efficient selection of usable pixels. Although efficient, the proposed method has its drawback in the necessity of parameter adjustments for particular sequence.

In most of the published work, to assess the problem of motion estimation errors, an error model and a priori knowledge about the high resolution image is used. In practice, modeling the errors and image characteristics had proved as difficult tasks to perform, resulting in poor errors suppression. Human observers, however, have no difficulties in spotting the artifacts in the image that are result of the erroneous motion estimation. An idea to exploit this fact using machine learning techniques is presented here.

The proposed novel approach achieves suppression of the influence of the outlier regions and the misaligned pixels of every LR image based on the visual appearance of the error artifacts. In the first step super-resolution is separately applied to the current image and every neighboring LR image. The artifacts in every SR image are identified using learning-based algorithm and weights are created for every LR image containing the information about the usability of the LR-pixels. Using the weights and the LR images a new, final SR image is created that is free of error artifacts. The robustness of the proposed algorithm is evaluated experimentally and the results are presented here.

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2. PROPOSED ALGORITHM

2.1. The algorithm structure

The algorithm structure is illustrated in Fig. 1. In the Phase 1 of the algorithm a weight matrix is created for every neighboring LR image $\mathbf{g}_i$, i=1,…,N. An SR image is estimated using the current LR image $\mathbf{g}_c$ (the one to be enhanced) and only one of the neighboring LR images (SR in Fig.1). The iterative L_2-norm minimization was used as SR process, as described in Subsection 2.2. Three iterations have been empirically chosen, as a small number at which the artifacts can be successfully identified. The presence of the artifacts in every pixel of the SR image $\mathbf{f}_{ei}$ is than estimated (AD in Fig. 1) using machine learning based algorithm, in order to create weight matrix $\mathbf{M}_i$. In the Phase 2, the final super-resolution (FSR in Fig.1) is performed using all neighboring LR images $\mathbf{g}_i$ and the associated weight matrices $\mathbf{M}_i$ containing the information about the usability of the LR pixels, and the current image $\mathbf{g}_c$.

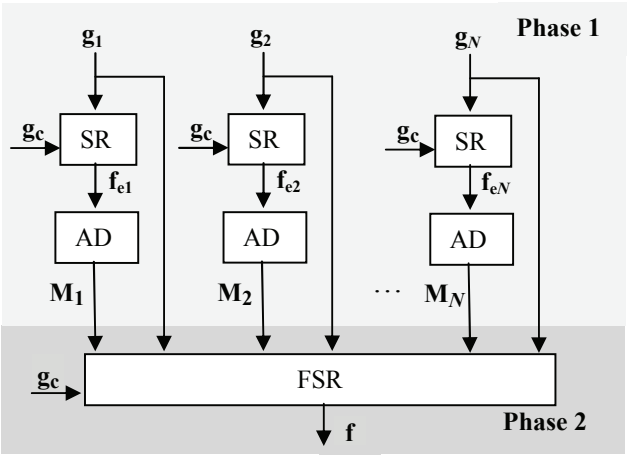


Fig.1. Block diagram of the proposed algorithm.

2.2. SR algorithm

The approach we are proposing could be applied with variety of SR algorithms. Here, to demonstrate the potential of the approach, we use L_2-norm minimization, as it was used in [2]. The registration is performed prior to SR process.

In order to estimate HR image $\mathbf{f}$, the following minimization formula was used:

$$\hat{\mathbf{f}} = \underset{\mathbf{f}}{\operatorname{argmin}} \left[\sum_{i=1}^{N} \left\| \mathbf{W}_i \mathbf{f} - \mathbf{g}_i \right\|_2^2 + \lambda \left\| \mathbf{Q}\mathbf{f} \right\|_2^2 \right] \qquad (1)$$

where $\mathbf{W}_i$ represents warping, blurring and down sampling of ith LR image, $\mathbf{Q}$ represents the a priori knowledge about the image and λ is regularisation coefficient. The minimization of the cost function is performed using gradient descent method. At every iteration, the solution is updated as follows:

$$\hat{\mathbf{f}}_{j+1} = \hat{\mathbf{f}}_j + \boldsymbol{\mu} \left[\sum_{i=1}^{N} \mathbf{W}_i^T \left[\mathbf{W}_i \hat{\mathbf{f}}_j - \mathbf{g}_i \right] + \lambda \mathbf{Q}^T \mathbf{Q} \hat{\mathbf{f}}_j \right] \qquad (2)$$

In the SR process of the Phase 1 no regularisation was used (λ=0) in order to bust the error artifacts and, by that, make them easy for detection. The initial estimate, $\hat{\mathbf{f}}_0$, is the bicubic transform of $\mathbf{g}_c$.

In order to incorporate the information about the pixel usability estimated in Phase 1, in the final super-resolution process (FSR in Phase 2, Fig. 1), the above equation is modified similar to the modification proposed in [2]:

$$\hat{\mathbf{f}}_{j+1} = \hat{\mathbf{f}}_j + \boldsymbol{\mu} \left[\sum_{i=1}^{N} \mathbf{M}_i \mathbf{W}_i^T \left[\mathbf{W}_i \hat{\mathbf{f}}_j - \mathbf{g}_i \right] + \lambda \mathbf{Q}^T \mathbf{Q} \hat{\mathbf{f}}_j \right] \qquad (3)$$

where $\mathbf{M}_i$ is weight matrix for the ith LR image. Here, differently from [2], the weight coefficients are real numbers in the range [0, 1], calculated as described in Subsection 2.3.5.

It should be noticed that the SR process in the Phase 1 (used to estimate SR images for the detection of the artifacts) and FSR process in the Phase 2 (used for estimation of the final SR image) need not to be same. For the first SR process a method that is fast, but sensitive to registration errors could be used. In that way the computational complexity would be lower and registration error artifacts more pronounced and, by that, easier to detect. As a second SR process (FSR) a more complex method that has better sharpening capabilities could be used.

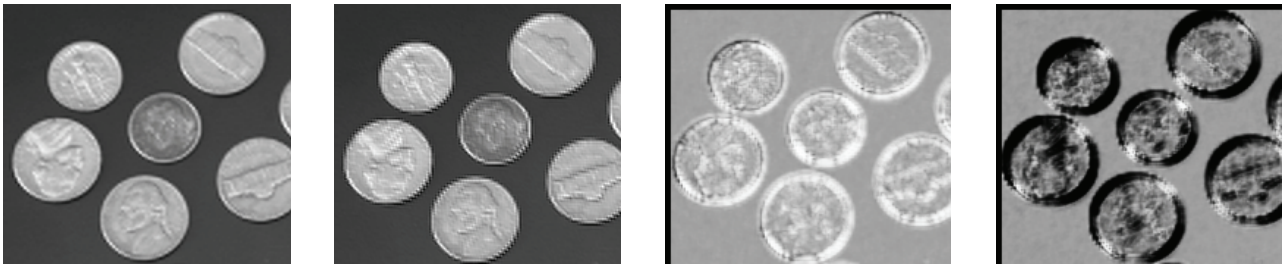
2.3. Artifact detection

The artifacts caused by misregistration of the LR images are detected using machine learning algorithm. The algorithm outputs the probability (further used as weight coefficient) the pixel being tested and its close surrounding to be part of visible artifacts. In this Subsection, the form of the artifacts, the training of the detection algorithm and the calculation of the weight matrix are described.

2.3.1. Artifacts description

The error in the registration of the LR images manifests as unpleasing artifacts in the SR image. Fig.2.b) shows the SR image obtained using erroneous motion vectors where the error is spatially invariant i.e. it is constant for every pixel, regardless of the position of the pixel in the image. The magnitude of the error is 1 pixel in the direction represented with the arrow on Fig. 2.b). Artifacts occur as a “texture” region around the edges of the objects in the SR image, similar to the well known ringing effect. The type of the “texture” and the intensity of the artifacts depend on the size

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a) b) c) d)

Fig. 2. Artifact detection: a) Result of the SR (without errors in the registration), b) Result of the SR (with errors in the registration). The arrow shows the direction of the error, c) Result from the artifact detection for the image shown on Fig. 2.a), d) Result from the artifact detection for the image shown on Fig. 2.b)

and the direction of the error vector. The artifacts are most intensive for edges with a direction perpendicular to the direction of the error. However, they are visually most unpleasing for edges covering an angle of 45 degrees with the direction of the error.

2.3.2. Feature descriptor

The feature descriptor used for the detection of the artifacts is extracted using image block of size 7x7 pixels, centered at the pixel location that is tested for artifacts. The artifacts, which occur as a result of misalignment of the LR images, usually, have high frequency components, therefore the images are analyzed in frequency domain.

The feature vector consists of two parts. The first part contains information about what has been added to the bicubic transform in the SR process and the second part contains information about the bicubic transform, itself. The task of the algorithm is to distinguish the pairs of information (the first and the second part of the descriptor) who correspond to improvement of the image from the pairs who correspond to appearance of artifacts.

The first part is obtained in the following way. Two 7x7 blocks are extracted, one from the SR image and the other from the upscaled current LR image generated using bicubic interpolation. The blocks are subtracted and the result is multiplied with a Gaussian weighting function with standard deviation 0.4, positioned at the center of the block, in order to lower the influence of the outer pixels. Next, the 2D DFT is applied and a half of coefficients of the amplitude spectrum are raster scanned and used as the first 25 coefficients of the feature vector.

The second part of the feature vector is obtained from a 7x7 block extracted from the upscaled current LR image generated using bicubic interpolation and filtered using 3x3 Laplace filter. Again, the result is multiplied with a Gaussian weighting function with standard deviation 0.4, positioned at the center of the block, and the 2D DFT is applied. One half of the coefficients of the amplitude spectrum is used as a second part of the feature vector, thus the dimensionality of the feature vector is 50.

2.3.3. First step of the training

It is expected that the coefficients in the first part of the feature vector (the first 25 coefficients) from an image that does not contain artifacts or outlier regions, are highly correlated to the coefficients in the second part of the feature vector (the last 25 coefficients). On the other hand, if the feature vector is from an image that contains artifacts, the correlation of coefficients will be lower. The correlation in the feature vector is used as cue that indicates the presence of artifacts or outlier regions.

Principal component analysis (PCA) was used to emphasize the difference in correlation of the coefficients in a feature vectors from error-free and erroneous images. The PCA is performed on the feature vectors from the error-free images and, as a result, a transform matrix **P** is obtained. When the transform matrix **P** is applied to the feature vector from an image that does not contain artifacts, the energy of the transformed signal will be contained in a small number of coefficients. If the feature vector is from an image that contains artifacts the energy of the signal will be scattered in a larger number of coefficients. As a result of the application of PCA the obtained feature vectors are easier to classify.

In order to avoid problems with matrix singularity in the training process, the last 6 coefficients of the transformed feature vector are rejected. The final feature vector consists of 44 coefficients.

2.3.4. Second step of the training

As a machine learning method, logistic regression, was used. This method is relatively simple and outputs continuous values. The hypothesis of the learning algorithm is:

$$h_\theta(x) = \frac{1}{1+e^{-\theta^T x}}, \qquad (4)$$

where **x** is the transformed feature vector and θ are the parameters of the model.

The probability, the pixel of the image to belong to one of the classes, knowing the feature vector and parameters θ, is modeled by:

$$p(y|x;\theta) = (h_\theta(x))^y (1 - h_\theta(x))^{1-y}, \qquad (5)$$

where y is 0 if the feature vector is from an image that contains artifacts (first class) and 1 if feature vector is from

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an image that does not contains artifacts (second class).
In order to estimate the parameters $\boldsymbol{\theta}$, the likelihood function of the parameters can be written as:

$$L(\theta) = \prod_{i=1}^{M} p(y_i|x_i;\theta)$$

$$= \prod_{i=1}^{M} (h_\theta(x_i))^{y_i}(1-h_\theta(x_i))^{1-y_i} \quad (6)$$

The parameters are obtained by maximizing the log-likelihood function $l(\theta) = logL(\theta)$, using the gradient ascent method.

2.3.5. Calculation of the weights
For a particular SR image $\mathbf{f}_{ei}$, i=1, …, *N*, a feature vector is extracted for every pixel location (*r*,*c*) in the image (as described in Subsection 2.3.2) and the transform using transform matrix **P** is applied, resulting in transformed feature vector *x*(*r*,*c*). A weight coefficient is calculated using the probability function in (4), i.e. $\mathbf{M}_i(r,c)=h_\theta(\mathbf{x}(r,c))$. Fig. 2.d) shows the results of the calculation of the weights for a frame with spatially invariant error in the motion vectors and Fig. 2.c) shows the weights for a super-resolved image without errors in the registration parameters, white designating weight value 1 (no visible artifacts present) and black designating weight value 0 (severe artifacts present). As can be observed from Fig. 2.c), although there is no registration error, some dark regions are present. They are mostly located around the edges of coins where there is visible ringing effect as a result of SR process. This is to be expected, since the visual appearance of the ringing effect is very similar to the appearance of the registration error artifacts, and it shoes that the proposed algorithm will also suppress ringing artifacts of different origin increasing the visual quality of the image. Fig. 2.d) shows that the error artifacts are detected where they are most annoying (in the vicinity of edges that are perpendicular to registration error). The areas with edges not affected by registration error have high weights values, allowing for effective SR.

2.3.6. Training set
For the training of the algorithm, 42 different grayscale images were used. In order to obtain feature vectors from SR images that do not contain artifacts, two different LR images are created from every training image, one using zero motion and the other using arbitrary motion vectors and SR is performed using the two images. Since the accurate motion vectors are known the resulting image is free of artifacts caused by misalignment of the LR images. Next, the edges of the bicubicly interpolated LR image are detected using the Canny-method. In general, Canny operator produces relatively large number of edge pixels. Using all edge pixels to produce feature vectors would result in a huge number of training vectors. In order to lower that number and assuming that error artifacts in the vicinity of neighboring edge pixels are similar, only edge pixels that are positioned on the 15 pixels grid in both directions are selected. Since the error artifacts often spread few pixels around the edges, a feature vector is extracted for every pixel in a 7x7 neighborhood centered on selected edge pixels.
In order to get training feature vectors from images that contain artifacts from misaligned LR images, the same procedure is repeated with addition of arbitrary error to the motion vectors. For the first 31 images the error in the motion vector is spatially variant, with uniform distribution in the range of 0.5 to 5 pixels. For the remaining 11 images the error of the motion vectors is spatially invariant, with uniform distribution in the range 0.4 to 1.5 pixels. In order to simulate the presence of outliers, the DC level of 5 training SR images was changed and for 5 other training images the feature vector was calculated using different bicubically upscaled LR image.

3. RESULTS

To test the performance of the proposed algorithm with the real video sequences, the "Football" sequence was chosen as a very fast changing sequence in which a lot of outliers and motion estimation error are to be expected. In order to test the algorithms capabilities to effectively suppress error artifacts, no regularization term was used in the SR process. Results from application of different SR methods are presented in Fig. 3. The size of the images is first lowered by a factor of two in both directions, using nonoverlapping averaging, and registration is performed using quarter pixel full search block matching algorithm. The 245th frame is super-resolved using different number of neighboring frames. Fig 3.e) shows the outcome of the proposed algorithm with the use of 6 neighboring frames. For comparison, Figs. 3.b), c) and d) show the outcome of the algorithms proposed in [1], [7] and [2], respectively. It can be seen that the SR image obtained with the proposed algorithm has same sharpness as the ones obtained with the algorithms in [7] and [2], and higher sharpness than the ones obtained using bicubic interpolation and the algorithm in [1]. Also, the proposed algorithm results in SR image that is artifacts-free, while the algorithms in [7] and [2] produce SR images with visible artifacts encircled with ellipses in Fig. 3.c) and d). It can be concluded that the proposed algorithm suppresses the artifacts without loss of useful information. Fig 3.f) shows the PSNR values as a function of the number of LR frames used in the SR procedure for different SR algorithms. It can be seen that PSNR values for the algorithms in [1], [2] and [7] drop abruptly when the number of frames is 3 due to introduction of artifacts. The PSNR value decreases with the number of included frames. The proposed algorithm does not suffer from these effects and maintains highest PSNR value. Fig. 4.b) shows the

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Fig. 3.a) bicubic interpolation



Fig. 3.b) SR algorithm proposed in [1]



Fig. 3.c) SR algorithm proposed in [7]



Fig. 3.d) SR algorithm proposed in [2]



Fig. 3.e) The proposed algorithm

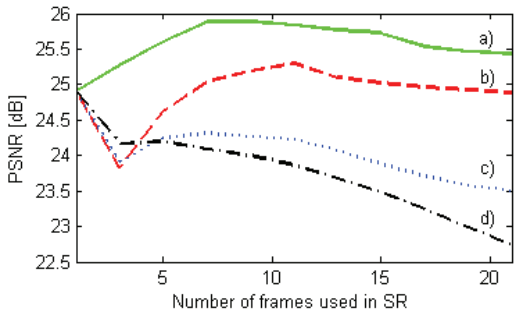


Fig. 3.f) PSNR curves: a) proposed algorithm, b) SR algorithm proposed in [2], c) SR algorithm proposed in [1], d) SR algorithm proposed in [7]

outcome of the proposed algorithm for the frame 113 of the "Bus" sequence. Frames 111-115 of the sequence were used as LR images, as it was done in [6]. The SR Fig.4c)-f) show the results taken from [6]. Although the used frames are the same, the direct comparison of the algorithms outcome is not possible due to possibly different motion vectors and image resolution. In order to avoid possible misunderstanding in favor of the proposed algorithm, the images are shown with different size. Nevertheless, a comparison can be performed focusing on the occurrence of artifacts. It can be seen that the result from the proposed algorithm is without artifacts, while all other results contain visible artifacts.

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Fig. 4.a)bicubic transform



Fig. 4.b) Our result

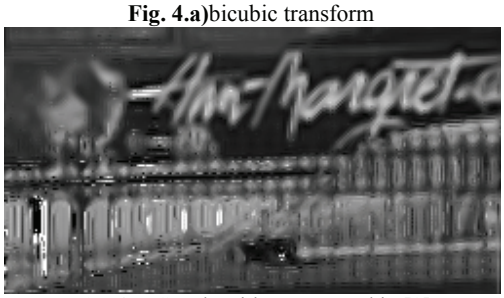


Fig. 4.c) SR algorithm proposed in [8]

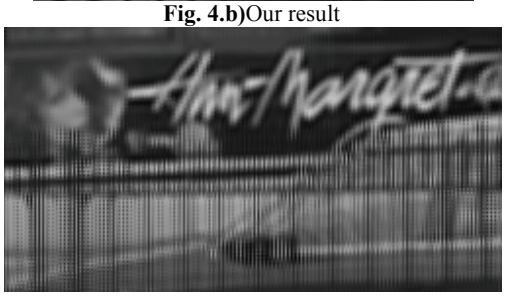


Fig. 4.d) SR algorithm proposed in [9]

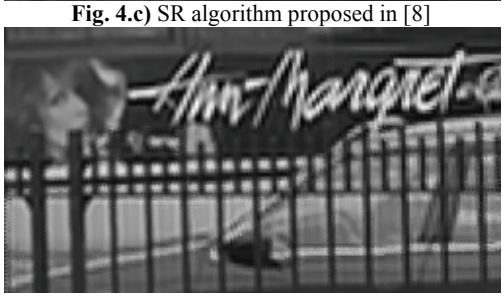


Fig. 4.e) SR algorithm proposed in [6]

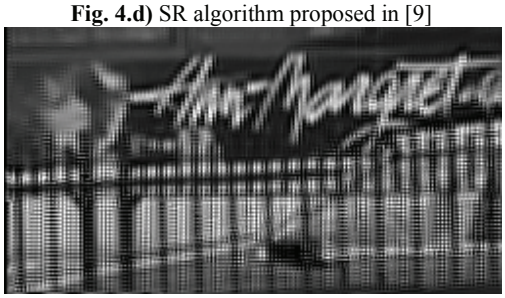


Fig. 4.f) SR algorithm proposed in [10]

4. CONCLUSION

The paper presents a novel approach to the problem of multiple frame super-resolution in the presence of registration errors and outliers. The proposed approach is based on machine learning in order to detect introduction of error artifacts in the SR image. The information about the indentified error artifacts is used to generate appropriate weight matrices for artifacts suppression. The experimental results obtained using L_2-norm minimization show that the proposed approach is very effective in error artifacts suppression, while retains SR algorithms sharpening capabilities. Furthermore, it can be used in combination with different SR methods or regularizations, possibly yielding even better results.

5. REFERENCES

[1] S. Farsiu, D. Robinson, M. Elad, and P. Milanfar, Fast and robust super-resolution," Proc. 2003 IEEE Conf. Image Processing 2, pp. 291-294, 2003.

[2] Z.Ivanovski, Lj.Panovski and L.J.Karam, "Robust super-resolution based on pixel-level selectivity", VCIP 2006.

[3] S. Farsiu, D. Robinson, M. Elad, P. Milanfar, "Fast and Robust Multi-frame Super-resolution", IEEE Trans. Image Proc., vol. 13, no. 10, pp. 1327-1344, October 2004.

[4] N. A. El-Yamany and P. E. Papamichalis, "An adaptive M-estimation framework for robust image superresolution without regularization," in Visual Comm. Image Proc., vol. 6822, Proc. of SPIE, pp. 1–12, San Jose, USA, Jan. 2008.

[5] H.Takeda, P.Milanfar, M.Protter, M.Elad, "Super-resolution without explicit subpixel motion estimation," IEEETrans. Image Proc., vol.18,no.9,pp.1958-75, Sept 2009

[6] S.P. Belekos, N.P. Galatsanos, and A.K. Katsaggelos, "Maximum a posteriori video super-resolution with a new multichannel image prior," in Proc. EUSIPCO 2008, Lausanne, Switzerland, August 25-29, 2008.

[7] R. C. Hardie, K. J. Barnard, and E. E. Armstrong, "Joint MAP registration and high-resolution image estimation using a sequence of undersampled images," IEEE Trans. on Image Processing, vol. 6, no. 12, pp. 1621-1633, 1997

[8] D. Keren, S. Peleg, and R. Brada, "Image sequence enhancement using sub-pixel displacements," in Proc. IEEE Int. Conf. Computer Vision and Pattern Recog. (CVPR'88), Ann Arbor, Mich, USA, Jun 1988.

[9] P. Vandewalle, S. E. Susstrunk, and M. Vetterli, "A frequency domain approach to registration of aliased images with application to superresolution," EURASIP Journal on Applied Signal Proc., vol. 2006, pp. 1–14, 2006.

[10] M.Irani, S.Peleg, "Motion analysis for image enhancement: Resolution, occlusion, and transparency," Journal of Visual Comm. Image Rep., vol.4, no.4, pp. 1993–12, 1993.

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